

LAPORAN PRAKTIKUM MODUL 10

Modern JavaScript (ES6+)

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Mata Kuliah: Pemrograman Web

S1 Terapan Sistem Informasi Kota Cerdas

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2026

1.1 TUJUAN

- Memahami konsep dasar JavaScript sebagai bahasa pemrograman web.
- Memahami fitur Modern JavaScript (ES6+) dan perannya sebagai fondasi React.
- Mampu menggunakan sintaks ES6+ secara tepat dan konsisten.
- Memahami konsep state, immutability, dan functional programming.
- Mampu membangun aplikasi sederhana berbasis state menggunakan Vanilla JavaScript.

1.2 ALAT DAN BAHAN

- Komputer atau laptop (Windows/Linux/macOS)
- Browser modern (Google Chrome / Mozilla Firefox / Microsoft Edge)
- Code Editor (Visual Studio Code)
- Ekstensi Live Server (VSCoDe)
- File HTML dasar
- File JavaScript (.js)

1.3 LANGKAH-LANGKAH PRAKTIKUM

- Menyiapkan struktur folder dan file proyek webstore
- Membuat file index.html sebagai entry point aplikasi
- Membuat file style.css untuk tampilan antarmuka
- Membuat file app.js sebagai entry point JavaScript (module)
- Membuat modul Store.js untuk manajemen state global
- Membuat modul utils.js berisi fungsi deepClone
- Membuat komponen ProductList.js untuk menampilkan daftar produk
- Membuat komponen Cart.js untuk manajemen keranjang belanja
- Membuat komponen Newsletter.js menggunakan jQuery
- Membuat productApi.js sebagai simulasi pengambilan data produk
- Menjalankan proyek menggunakan Live Server dan menguji fungsionalitas

1.4 HASIL DAN DOKUMENTASI

app.js



```
1  import { fetchProducts } from "../src/api/productApi.js";
2  import { getState, setState } from "../src/core/Store.js";
3  import { ProductList } from "../src/components/ProductList.js";
4  import { Cart } from "../src/components/Cart.js";
5  import { Newsletter } from "../src/components/Newsletter.js";
6
7  function addToCart(product) {
8      const state = getState();
9
10     setState({
11         cart: [...state.cart, product],
12     });
13
14     render();
15 }
16
17 function checkout() {
18     setState({ cart: [] });
19     alert("Thank you for your purchase!");
20     render();
21 }
22
23 function render() {
24     const state = getState();
25     ProductList(state.products, addToCart);
26     Cart(state.cart, checkout);
27 }
28
29 async function init() {
30     try {
31         const data = await fetchProducts();
32         setState({ products: data });
33         render();
34         Newsletter();
35     } catch (error) {
36         console.error("Failed to load products:", error);
37     }
38 }
39
40 init();
41
```

src/core/Store.js



```
1  import { deepClone } from "../utils.js";
2
3  let state = {
4    products: [],
5    cart: [],
6  };
7
8  export const getState = () => deepClone(state);
9
10 export const setState = (newState) => {
11
12   const nextState = { ...state, ...newState };
13
14   state = nextState;
15
16   console.log("[Store] State updated:", state);
17 }
```

src/core/utils.js



```
1  import { getState } from "../Store.js";
2
3  export function deepClone(obj) {
4    if (obj === null || typeof obj !== "object") {
5      return obj;
6    }
7
8    if (obj instanceof Date) {
9      return new Date(obj.getTime());
10   }
11
12   if (Array.isArray(obj)) {
13     return obj.map((item) => deepClone(item));
14   }
15
16   const clonedObj = {};
17   for (const key in obj) {
18     if (Object.prototype.hasOwnProperty.call(obj, key)) {
19       clonedObj[key] = deepClone(obj[key]);
20     }
21   }
22
23   return clonedObj;
24 }
```

src/api/productApi.js

```
1 export const fetchProducts = async () => {
2   return new Promise((resolve) => {
3
4     setTimeout(() => {
5       resolve([
6         {
7           id: 1,
8           title: "Mechanical Keyboard",
9           price: 150,
10          rating: { rate: 4.8, count: 230 },
11
12          image: "https://images.unsplash.com/photo-1618384887929-16ec33fab9ef?auto=format&fit=crop&w=800&q=80"
13        },
14        {
15          id: 2,
16          title: "Gaming Mouse",
17          price: 80,
18          rating: { rate: 4.6, count: 150 },
19
20          image: "https://images.unsplash.com/photo-1527814050087-3793815479db?auto=format&fit=crop&w=800&q=80"
21        },
22        {
23          id: 3,
24          title: "4K Monitor 24\"",
25          price: 300,
26          rating: { rate: 4.9, count: 85 },
27
28          image: "https://images.unsplash.com/photo-1527443224154-c4a3942d3acf?auto=format&fit=crop&w=800&q=80"
29        },
30        {
31          id: 4,
32          title: "Premium Coffee Beans",
33          price: 25,
34          rating: { rate: 4.7, count: 500 },
35
36          image: "https://images.unsplash.com/photo-1559056199-641a0ac8b55e?auto=format&fit=crop&w=800&q=80"
37        },
38        {
39          id: 5,
40          title: "Travel Tumbler",
41          price: 35,
42          rating: { rate: 4.5, count: 120 },
43
44          image: "https://images.unsplash.com/photo-1578500494198-246f612d3b3d?auto=format&fit=crop&w=800&q=80"
45        },
46        {
47          id: 6,
48          title: "Noise Cancelling Headphones",
49          price: 200,
50          rating: { rate: 4.8, count: 320 },
51
52          image: "https://images.unsplash.com/photo-1505740420928-5e560c06d30e?auto=format&fit=crop&w=800&q=80"
53        },
54      ]),
55    });
56  }, 800);
57 });
58 };
```

src/components/ProductList.js

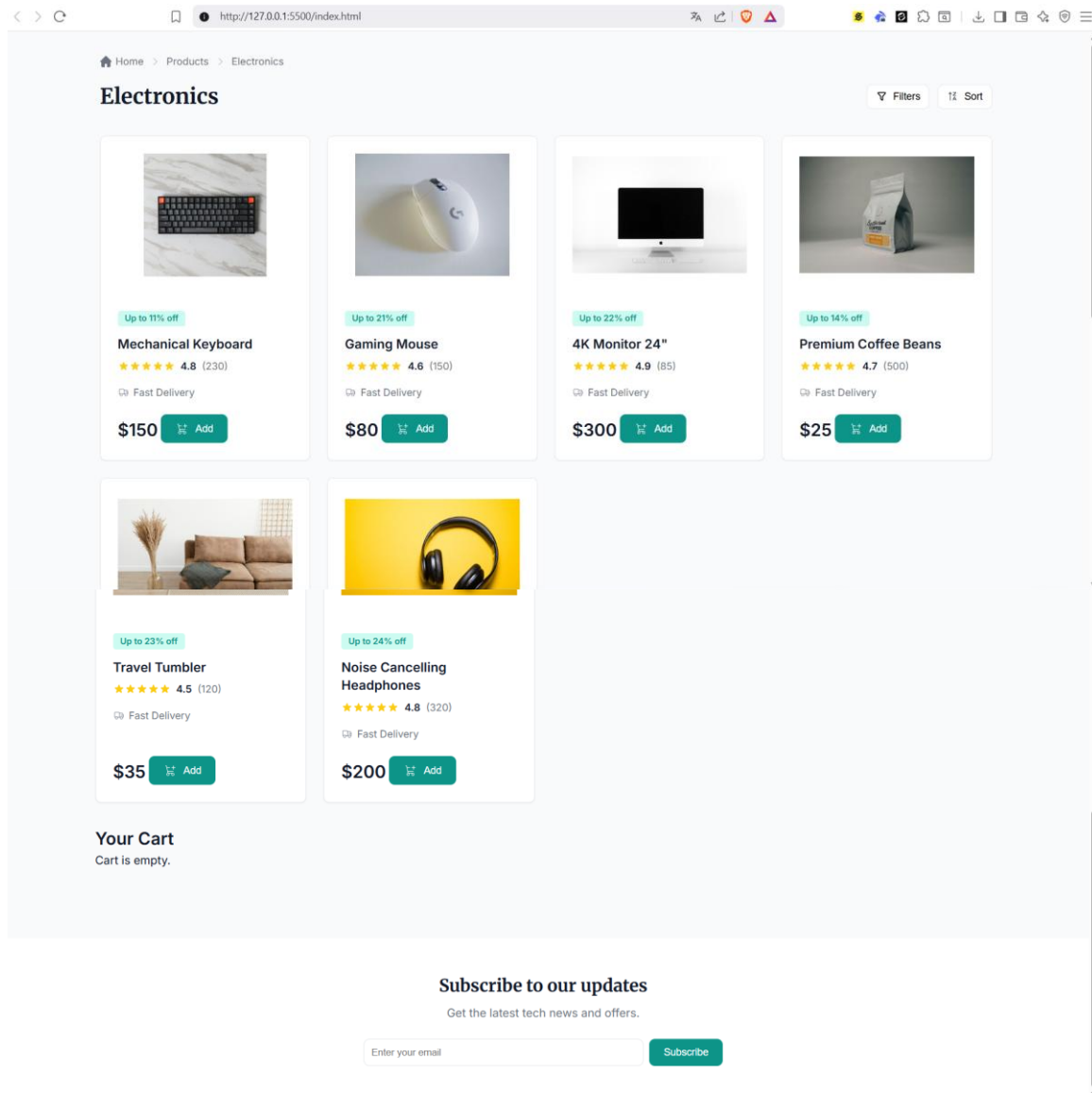
```
1  export function ProductList(products, onAdd) {
2  const container = document.getElementById("product-list");
3  container.innerHTML = "";
4
5  products.map((product) => {
6  const card = document.createElement("div");
7  card.className = "product-card";
8
9  const discount = Math.floor(Math.random() * 20) + 5;
10 const rating = product.rating || { rate: 4.5, count: 100 };
11
12 card.innerHTML = `
13 <div class="card-image">
14 
15 </div>
16 <div class="card-content">
17 <span class="badge">Up to ${discount}% off</span>
18 <a href="#" class="card-title" title="${product.title}">${product.title}</a>
19
20 <div class="rating-row">
21 <div class="stars">
22 ${getStarSVG().repeat(5)}
23 </div>
24 <span class="rating-score">${rating.rate}</span>
25 <span class="rating-count">${rating.count}</span>
26 </div>
27
28 <ul class="features-list">
29 <li class="feature">
30 <svg xmlns="http://www.w3.org/2000/svg" fill="none" viewBox="0 0 24 24">
31
32 <path stroke="currentColor" stroke-linecap="round" stroke-linejoin="round" stroke-
33 width="2" d="M13 7h6l2 4m-8-4v8m0-8V6a1 1 0 0 1-1H4a1 1 0 0 1-1v9h2m8 0H9m4
34
35 0h2m4 0h2v-4m0 0h-5m3.5 5.5a2.5 2.5 0 1 1-5 0 2.5 2.5 0 0 1 5 0Zm-10 0a2.5 2.5 0 1 1-5 0
36 2.5 2.5 0 0 1 5 0Z" />
37 </svg>
38 Fast Delivery
39 </li>
40 </ul>
41
42 <div class="card-footer">
43 <span class="price">${product.price}</span>
44 <button class="add-btn btn-primary">
45 <svg xmlns="http://www.w3.org/2000/svg" fill="none" viewBox="0 0 24 24">
46
47 <path stroke="currentColor" stroke-linecap="round" stroke-linejoin="round" stroke-
48 width="2" d="M4 4h1.5L8 16m0 0h8m-8 0a2 2 0 1 0 0 4 2 2 0 0 0 0-4Zm8 0a2 2 0 1 0 0 4 2 2
49 0 0 0 0-4Zm-.75-3H7.5M11 7H6.312M17 4v6m-3-3h6" />
50 </svg>
51 Add
52 </button>
53 </div>
54 </div>
55 `;
56
57
58 card.querySelector(".add-btn").addEventListener("click", function () {
59 onAdd(product);
60 });
61
62 container.appendChild(card);
63 });
64 }
65
66 function getStarSVG() {
67 return `
68 <svg xmlns="http://www.w3.org/2000/svg"
69 fill="currentColor"
70 viewBox="0 0 20 20"
71 width="20"
72 height="20">
73 <path d="M9.049 2.927c-.32-.921 1.603-.921 1.902 0l1.07 3.292a1
74 1 0 0 0.95.69h3.462c.969 0 1.371 1.24.588 1.811-2.8 2.034a1
75 1 0 0 0-.364 1.118l1.07 3.292c.3.921-.755 1.688-1.54
76 1.118-2.8 2.034a1 1 0 0 0-1.175 0l-2.8 2.034c-.784 1.811-.197-1.539-1.118 1.07-3.292a1
77 1 0 0 0-.364 1.118l2.98 8.722c-.783-.57-.38-1.811 1.811-1.81h3.461a1
78 1 0 0 0.951-.69l1.07-3.292"/>
79 </svg>`;
80 }
81 }
```

src/components/Cart.js

```
1  export function Cart(cart, onCheckout) {
2    const container = document.getElementById("cart");
3    container.innerHTML = "<h2>Your Cart</h2>";
4
5    if (cart.length === 0) {
6      container.innerHTML += "<p>Cart is empty.</p>";
7      return;
8    }
9
10   const list = document.createElement("div");
11   list.className = "cart-list";
12
13   cart.map((item) => {
14     const div = document.createElement("div");
15
16     div.className = "cart-item";
17
18     div.innerHTML = `
19       
20       <div>
21         <h4>${item.title}</h4>
22         <span>${item.price}</span>
23       </div>
24     `;
25
26     list.appendChild(div);
27   });
28
29   container.appendChild(list);
30
31   const total = cart.reduce((acc, item) => acc + item.price, 0);
32
33   const footer = document.createElement("div");
34
35   footer.innerHTML = `
36     <div class="total-row">
37       <span>Total:</span>
38       <span>${total.toFixed(2)}</span>
39     </div>
40
41     <button class="btn-primary" id="checkout-btn">
42       Buy Now
43     </button>
44   `;
45
46   footer.querySelector("#checkout-btn")
47     .addEventListener("click", onCheckout);
48
49   container.appendChild(footer);
50 }
51
```

src/components/Newsletter.js

```
1  export function Newsletter() {
2    const $container = $('<div class="newsletter-section"></div>');
3
4    const $content = $('
5      <div class="newsletter-content">
6        <h3>Subscribe to our updates</h3>
7        <p>Get the latest tech news and offers.</p>
8
9        <form id="newsletter-form">
10          <input type="email" placeholder="Enter your email" required />
11          <button type="submit" class="btn-primary">Subscribe</button>
12        </form>
13
14        <div class="success-message" style="display:none; margin-top: 10px; color: green;">
15          Thanks for subscribing!
16        </div>
17      </div>
18    `);
19
20    $container.append($content);
21    $("#app").append($container);
22
23    $content.find("form").on("submit", function (e) {
24      e.preventDefault();
25
26      const $form = $(this);
27
28      $form.slideUp(400, function () {
29        $content.find(".success-message").fadeIn();
30      });
31    });
32  }
33
```



1.5 ANALISIS DAN PEMBAHASAN

Pada praktikum ini, berhasil membangun aplikasi webstore sederhana menggunakan konsep-konsep Modern JavaScript (ES6+) yang menjadi fondasi pengembangan aplikasi berbasis React.

Secara keseluruhan, proyek webstore ini mengintegrasikan seluruh konsep ES6+ yang dipelajari selama praktikum ke dalam satu aplikasi nyata, mulai dari arrow function, destructuring, spread operator, module, callback, closure, hingga asynchronous JavaScript.

1.6 KESIMPULAN

Melalui praktikum ini, mahasiswa berhasil membangun aplikasi webstore sederhana menggunakan Vanilla JavaScript dengan menerapkan konsep-konsep Modern JavaScript Higher order function seperti `map()` sangat efektif untuk mengubah data array menjadi tampilan HTML secara dinamis tanpa memodifikasi data asli.

- jQuery dapat dikombinasikan dengan Vanilla JavaScript ES6+ untuk menyederhanakan manipulasi DOM dan animasi pada bagian tertentu dari aplikasi.
- Konsep-konsep yang dipelajari dalam praktikum ini merupakan fondasi langsung dari framework React yang akan dipelajari selanjutnya.

1.7 LAMPIRAN

Link repositori GitHub:

<https://github.com/AriqAhmadNayaka/WebPro4904/tree/FathirSyahragilYusuf707012500050>